Aleksei Piskunov

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Website: https://aleksei-piskunov.github.io/
Research interests: Algebraic geometry, moduli spaces of sheaves, geometric representation theory, mathematical physics.
Computer skills: LaTeX, Lean, Wolfram, Python (PyTorch, NumPy, pandas), C.

Education

2022–present **Technion – Israel Institute of Technology**, Haifa
Ph.D. in Mathematics, expected winter 2026–2027
Advisor: Howard Nuer
Thesis: Moduli spaces of stable sheaves of pure dimension one on bielliptic surfaces

2020–2022 **Higher School of Economics**, Moscow
Master's degree in Mathematics; Advisor: Constantin Shramov

2016–2020 **Higher School of Economics**, Moscow
Bachelor's degree in Mathematics; Advisor: Leonid Rybnikov

Research Papers

Preprints

- Topology and geometry of moduli spaces of semistable sheaves on bielliptic surfaces. arXiv:2607.18918.

In preparation

- Irreducibility of Moduli Spaces of Sheaves on Enriques and Bielliptic Surfaces. With Howard Nuer.
- Weak Brill-Noether on Enriques surfaces. With Howard Nuer, Edoardo Mason, Roberto Vacca, Huishi Yu, and Filippo Papallo.

Research Positions

2019–2022 Research Intern, Independent University of Moscow

Honors and Awards

2020–2025 Integrated Master–PhD Track Scholarship, Higher School of Economics, Moscow

Research Talks and Posters

2026 Moduli spaces of stable sheaves of pure dimension one on bielliptic surfaces
Technion Graduation Seminar

2026 Topology and geometry of moduli spaces of semistable sheaves on bielliptic surfaces
Technion–University of Haifa Joint Algebra Seminar

2026 Relative compactified Jacobians of linear systems on bielliptic surfaces (poster)
IIT Madras, Stacks and Moduli conference

2025 Relative compactified Jacobians of linear systems on bielliptic surfaces (poster)
Université de Toulouse, Conférence finale de l'ANR FanoHK

2025 Relative compactified Jacobians of linear systems on bielliptic surfaces
Stockholm University–KTH Joint Algebra Seminar (invited talk)

2022 Motivic zeta-function for surfaces and Severi-Brauer varieties
Technion–University of Haifa Joint Algebra Seminar

Teaching Experience

2023–2025 **Teaching Assistant, Technion – Israel Institute of Technology**
Algebraic Geometry (2024, 2025); Calculus I–II (2023–2024).

2019–2022 **Teaching Assistant, Independent University of Moscow**
Smooth Manifolds; Linear Algebra and Geometry; Algebra I–II; Calculus I.

2017–2021 **Teaching Assistant, Higher School of Economics, Moscow**
Lie Groups and Lie Algebras; Advanced Algebra; Algebra I–III; Geometry I.

2018–2022 **Private Tutor in Mathematics and Physics, Moscow**
Tutoring for school-leaving examinations and Olympiads, with additional support for university coursework.

2015–2018 **Physics Instructor, Maximum Education Center, Moscow**
Taught group classes preparing secondary school students for the Russian Unified State Exam in physics.

Academic Service

2023–2026 Co-organizer, Algebraic Geometry Research Group Learning Seminar, Technion

Spring 2025 Organizer, Reading Seminar on Abelian Varieties, Technion

Fall 2024 Organizer, Reading Seminar on Mixed Hodge Structures, Technion

Selected Expository Talks

Algebraic Geometry Research Group - Learning Seminar, Technion (Haifa)

2026 The theta divisor and the non-rationality of the cubic threefold

2025 Basic results of the Brill-Noether theory for curves

2024 Surfaces with $\kappa = 0$; Surfaces of general type

2023 Ruled surfaces

Representation theory seminar, Higher School of Economics (Moscow)

2026 Introduction to symmetric tensor categories

2025 Shifted Yangians, Nilpotent orbits and W-algebras

Iskovskikh's seminar, Steklov Mathematical Institute (Moscow)

2022 Symmetric powers of Severi-Brauer varieties

2021 Rationality criteria for motivic zeta functions of complex surfaces

Selected Conferences and Schools

Apr. 2026 Moduli Spaces in Various Flavors of Geometry, RIMS in Kyoto

Jan. 2026 Stacks and Moduli, IIT Madras, India

Dec. 2025 Conference on Fano and Hyperkähler Manifolds, Université de Toulouse, France

Nov. 2025 Workshop on Six-Functor Formalisms, TU Darmstadt, Germany

Jul. 2025 Summer Research Institute in Algebraic Geometry, Colorado State University, USA

May 2025 Conference on Representation Theory and Algebraic Geometry in Honor of Joseph Bernstein on the Occasion of His Birthday, Weizmann Institute of Science, Israel

Jul. 2023 Summer School on Algebraic Geometry: Derived Categories, Stability Conditions and Moduli Spaces, Technion, Israel